

CORE PURE / DIFFERENTIAL EQUATIONS

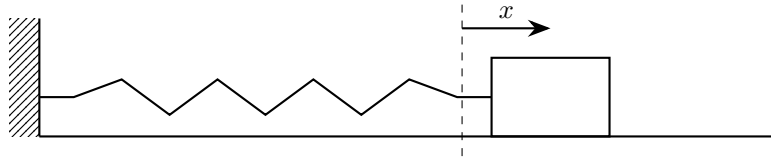
Modelling with 2nd Order Differential Equations

Questions by topic

12 questions · 160 marks

Questions

Question 1	14 marks	2
Question 2	23 marks	3
Question 3	13 marks	4
Question 4	5 marks	5
Question 5	22 marks	6
Question 6	14 marks	7
Question 7	13 marks	8
Question 8	12 marks	9
Question 9	7 marks	10
Question 10	14 marks	11
Question 11	12 marks	12
Question 12	11 marks	13



1. The horizontal displacement of a glider attached to a spring, shown in the diagram above, is modelled by the differential equation

$$\frac{d^2x}{dt^2} + 16x = 3 \sin 4t$$

where x is the displacement, in metres, of the glider from its equilibrium position, t seconds after the motion begins.

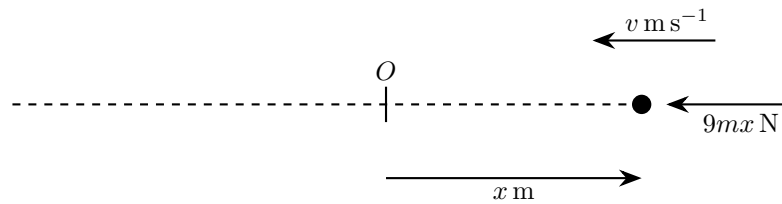
- (a) (i) Show that a particular solution of the differential equation is

$$x = -\frac{3}{8}t \cos 4t \quad [4]$$

- (ii) Hence, find the general solution of the differential equation [4]

Initially, the glider is 0.20 m to the right of equilibrium and is moving towards equilibrium at 0.50 m s^{-1} . Given that 4 seconds after the motion begins, the displacement of the glider is β m,

- (b) determine, according to the model, the value of β to 3 significant figures [4]
- (c) Given that the true value of β is 0.46 m, comment on the suitability of the model [1]
- (d) Write down a differential equation for the glider if the periodic driving force is removed, so that the motion becomes simple harmonic [1]



2. A particle of mass m kg moves in a horizontal line through its equilibrium position O . At time t seconds its displacement from O is x metres and its speed is v m s^{-1} . When the particle is to the right of O , it is acted on by a restoring force of magnitude $9mx$ newtons directed towards O , as shown in the diagram.

At $t = 0$, the particle is 1 m to the right of O and is projected towards O with speed 6 m s^{-1} .

- (a) In the absence of damping and any external force,

- (i) Show that the motion is modelled by the differential equation

$$\frac{d^2x}{dt^2} + 9x = 0 \quad [1]$$

- (ii) State the type of motion [1]
 (iii) Write down the period of the motion [1]
 (iv) Find x in terms of t [4]
 (v) Find the amplitude of the motion [2]

- (b) In a second model, the particle again starts from the same initial position and velocity, but the motion is damped by a force $6mv$ newtons acting opposite to the direction of motion.

- (i) Show that

$$\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 9x = 0 \quad [1]$$

- (ii) State, giving a reason, whether the system is under-damped, critically damped or over-damped [1]
 (iii) Determine the general solution of this differential equation [3]

- (c) In a third model, the particle again starts from the same initial position and velocity, and is subject both to the damping force in part (b) and to an external force $25m \cos 4t$ newtons acting in the positive direction. The motion is now modelled by the differential equation

$$\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 9x = 25 \cos 4t$$

- (i) Find x in terms of t [7]

- (ii) In the long term, the particle is observed to perform simple harmonic motion with period about 1.6 seconds. Verify that this is consistent with the answer to part (c)(i) [2]

- 3.** The charge q on a capacitor in a driven RLC circuit is modelled by

$$\frac{d^2q}{dt^2} + 2\frac{dq}{dt} + 10q = 37 \cos 3t$$

where $t \geq 0$ is time.

When $t = 0$, $q = 1$ and $\frac{dq}{dt} = 15$.

(a) Find q as a function of t [9]

(b) Hence write down an approximate expression for q when t is large, giving your answer in the form $R \cos(3t - \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$ [4]

4. The vertical displacement of the body of a car from its steady running height is measured by y metres, where y is positive upwards. The body can move above or below this level, so y can take positive and negative values. After the car leaves a speed hump, the body is momentarily above its steady running height and at rest at time $t = 0$.

A proposed model for the subsequent motion, for $t \geq 0$, is

$$4 \frac{d^2 y}{dt^2} + p \frac{dy}{dt} + 9y = 0,$$

where p is a constant and $p \geq 0$.

- (a) (i) According to the model, for what value of p would the motion be simple harmonic? [1]
(ii) Explain briefly why modelling the motion as simple harmonic is unlikely to be realistic for a car suspension [1]
- (b) Find the range of values of p for which the model predicts that the body of the car never goes below its steady running height [2]
- (c) Sketch a possible graph of y against t when p lies outside the range found in part (b) but the motion is not simple harmonic [1]

5. A load of mass m kg hangs from a light vertical spring and is free to move only vertically in a liquid.

The spring has spring constant $k \text{ N m}^{-1}$, and the liquid exerts a resistance force of magnitude cv newtons opposite to the direction of motion, where $v \text{ m s}^{-1}$ is the speed of the load.

Let y metres be the downward displacement of the load from its equilibrium position at time t seconds. Assume that $4km > c^2$.

- (a) Show that the motion is modelled by

$$m \frac{d^2 y}{dt^2} + c \frac{dy}{dt} + ky = 0$$

and hence show that

$$y = e^{-\frac{ct}{2m}} \left(P \cos \left(\frac{\sqrt{4km - c^2}}{2m} t \right) + Q \sin \left(\frac{\sqrt{4km - c^2}}{2m} t \right) \right), \quad [7]$$

where P and Q are constants.

- (b) It is given that

$$\begin{aligned} m &= 5 \\ c &= 30 \\ k &= 125 \end{aligned}$$

At time $t = 0$, the load is 0.12 m below its equilibrium position and is moving upwards with speed 0.12 m s^{-1} .

- (i) Find the values of P and Q [4]

- (ii) Find the first positive value of t for which the load is at its equilibrium position [5]

- (c) A second model ignores the resistance force but keeps the same values of m and k . Using the same initial conditions as in part (b), find an expression for y in terms of t , giving numerical constants to two significant figures where appropriate [6]

6. A manufacturer is testing the response of an electronic weighing platform. The displayed mass, m , measured in kilograms, after a box is placed on the platform is modelled by the differential equation

$$4 \frac{d^2m}{dt^2} + 4 \frac{dm}{dt} + 5m = 20$$

where t is the number of seconds after the box is placed on the platform.

- (a) Find a general solution of the differential equation [4]

Initially, the display reads 0 kg and the displayed mass is increasing at 5 kg s^{-1} .

- (b) Find, according to the model, the greatest value shown on the display [8]

The machine is programmed to print a label only when the displayed mass is less than 4.05 kg.

- (c) Determine whether, according to the model, the machine will print a label exactly 8 seconds after the box is placed on the platform [2]

7. An analogue meter has a damped pointer. After a short electrical pulse, the angular displacement θ radians of the pointer from its rest position, measured positive clockwise, is modelled by

$$\frac{d^2\theta}{dt^2} + 5\frac{d\theta}{dt} + 6\theta = 4e^{-4t}$$

where t is the time, in seconds, after the motion begins.

In one test, at $t = 0$ the pointer is passing through its rest position and has angular velocity 1 rad s^{-1} anticlockwise.

For this test, use the model to

- (a) find an equation for θ in terms of t [8]

- (b) find, to 3 decimal places, the greatest clockwise angular displacement of the pointer from its rest position [4]

In this test, the time taken for the pointer to pass through its rest position again was measured as 0.69 seconds.

- (c) State, with justification, whether or not this supports the model [1]

8. A trolley moves on a horizontal track and is attached to a spring and a dashpot. The trolley is pulled to the right of its equilibrium position and released.

At time t seconds after release, its horizontal displacement is x metres to the right of a fixed point P , where P is the equilibrium position and is 1.2 metres from the left-hand end of the track.

When $t = 0$:

$$x = 0.18,$$

the velocity of the trolley is 0 m s^{-1} ,

the acceleration of the trolley is -0.10 m s^{-2}

In the subsequent motion, the displacement of the trolley is modelled by the differential equation

$$3k \frac{d^2x}{dt^2} + 2k \frac{dx}{dt} + 5x = 0, \quad t \geq 0,$$

where k is a positive constant.

- (a) Find the value of k [2]
- (b) Find the particular solution of the differential equation [7]
- (c) Hence find, according to the model, the distance of the trolley from the left-hand end of the track 4.5 seconds after it is released [2]
- (d) State one limitation of the model [1]

9. In a feedback system, the error signal E is measured in suitable units.

For $t \geq 0$, E is modelled by the differential equation

$$\frac{d^2E}{dt^2} + \frac{dE}{dt} - 6E = -12$$

Initially the error is decreasing at 9 units per second, so $\frac{dE}{dt} = -9$ when $t = 0$. It is given that E tends to a finite limit as $t \rightarrow \infty$.

(a) Determine the value of E when $t = \ln 2$ [6]

(b) Determine the limit of E as $t \rightarrow \infty$ [1]

- 10.** A machine part executes damped forced vibrations. Its displacement y centimetres from equilibrium at time t seconds is modelled by

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 10y = 20\cos t + 5\sin t, \quad t \geq 0$$

- (a) Find the general solution of this differential equation. [6]
- (b) Given that $y = 3$ and $\frac{dy}{dt} = 3$ when $t = 0$, find the particular solution describing the motion. [4]
- (c) The first turning point of this motion for $t > 0$ occurs when $y = a$. Find a correct to 3 significant figures. [4]

11. A particle Q moves along the x -axis under a single force. When the time is t seconds, the displacement of Q from the origin is x metres and its velocity is $v \text{ m s}^{-1}$.

The motion of Q is modelled by the differential equation

$$\ddot{x} + \lambda^2 x = 0$$

where $\lambda \text{ rad s}^{-1}$ is a positive constant.

At time $t = 0$, Q is at the point A where $x = -\frac{1}{2}a$, for some positive constant a , and is moving in the positive direction.

Later, Q reaches the point B where $x = a$ and comes to instantaneous rest.

- (a) Write down the general solution of this differential equation. [1]

- (b) Using the answer to part (a), determine expressions, in terms of λ , a and t only, for the following quantities:

- (i) x [3]
(ii) v

- (c) Hence show that

$$v^2 = \lambda^2(a^2 - x^2) \quad [1]$$

- (d) The quantity r is defined by $r = \frac{1}{v}$. Using part (c), determine an expression for r_m , the mean value of r with respect to the displacement, as Q moves directly from A to B . [4]

- (e) One measure of the validity of the model is consideration of the value of r_m . If r_m exceeds 7.2 then the model is considered to be valid. The value of a is measured as 0.240 to 3 significant figures. The value of λ is measured as 0.80 ± 0.02 . Determine what can be inferred about the validity of the model from the given information. [1]

- (f) Find, according to the model, the least possible value of the velocity of Q at time $t = 0$. Give your answer to 2 significant figures. [2]

- 12.** An automated inspection trolley of mass 2 kg moves along a straight horizontal rail. The point A is fixed on the rail. The displacement of the trolley from A at time t seconds is x metres, for $t \geq 0$.

For the interval considered, the trolley stays to the right of A and continues to move away from A .

For $t \geq 0$, the trolley is subject to three forces modelled as follows:

- A driving force of magnitude $4(t + 1)$ N acts in the positive x -direction.
- A magnetic force of magnitude x N acts towards A .
- A resistive force of magnitude $2 \left| \frac{dx}{dt} \right|$ N acts opposite to the direction of motion.

- (a) Applying Newton's second law gives the differential equation

$$4(t + 1) - x - 2 \frac{dx}{dt} = 2 \frac{d^2x}{dt^2}$$

- (i) Explain the sign of the term involving $\frac{dx}{dt}$ in this differential equation [1]
- (ii) When $t = 0$ the displacement of the trolley is 1 m, and it is moving away from A with speed 1 m s^{-1} . Without attempting to solve the differential equation, find the acceleration of the trolley when $t = 0$ [2]
- (b) Let the particular solution to the differential equation in part (a) be a function g such that $x = g(t)$ for $t \geq 0$. The particular solution can be expressed as a Maclaurin series.
- (i) Show that the Maclaurin series for $g(t)$ up to and including the term in t is $1 + t$ [1]
- (ii) Use your answer to part (a)(ii) to show that the term in t^2 in the Maclaurin series for $g(t)$ is $\frac{1}{4}t^2$ [1]
- (iii) By differentiating the differential equation in part (a) with respect to t , show that the term in t^3 in the Maclaurin series for $g(t)$ is $\frac{1}{6}t^3$ [4]
- (c) You are given that the complete Maclaurin series for the function g is valid for all $t \geq 0$. After 0.4 seconds a sensor records that the trolley is 1.45 m from A .
- (i) Using the Maclaurin series for $g(t)$ up to and including the term in t^3 , evaluate the suitability of the model for determining the displacement of the trolley from A when $t = 0.4$ [1]
- (ii) Explain why it would not be sensible to use the Maclaurin series for $g(t)$ up to and including the term in t^3 to evaluate the suitability of the model for determining the displacement of the trolley from A when $t = 8$ [1]